

Kill Two Stacks with One Stone: “Fitts” Yourself while Doing Rehabilitation

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ABSTRACT

In this work, we describe our experience testing the potential dual role of rehabilitation exergames: as a therapeutic tool and as a means of studying Fitts’ Law. We initially created a game to encourage head movements for cervical rehabilitation. However, we soon recognized its potential to expand beyond its original purpose due to its resemblance to the ISO one-directional test based on Fitts’ paradigm. Our investigation delves into the impact of two apparatuses (exergame and one-directional test) on throughput, movement time, and error rate; analyzing also the effect of the sex factor. We present a novel integration of rehabilitation and Fitts’ law evaluation to provoke the HCI community to look beyond the mere pursuit of technological goals and to exploit every opportunity to deepen into the study of human behavior: from an exergame as a bridge to rehabilitation, to a HCI tool.

CCS CONCEPTS

• **Human-centered computing** → Empirical studies in HCI; • **Applied computing** → Health care information systems.

KEYWORDS

Fitts’ Law, head-tracker interface, Throughput, ISO 9241-411, cervical rehabilitation, exergame

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1 INTRODUCTION

In Human-Computer Interaction (HCI), the search for efficient, engaging, and adaptable interfaces continues to be a main point of research and innovation. As technology advances and our understanding of user behavior deepens, new opportunities arise to explore unconventional applications of established HCI principles. This study originated in the domain of rehabilitation exergames and then transcended its initial boundaries to test the foundational concepts of Fitts’ Law within this context.

Our work started within the domain of health and rehabilitation, where the integration of technology has given rise to the field known as “health tech” or digital health. Usually, the highway to integrate technology for rehabilitation purposes is to provide a health oriented application to a previously developed and evaluated (from to point of view of HCI) interface. In this work, we go further, and we explain how, from an application developed for rehabilitation purposes, we saw the opportunity to use it also for HCI evaluation, closing the cycle. And, therefore, killing two birds with one stone.

One notable innovation in the field of digital health is the development of rehabilitation exergames (interactive, motion-based activities designed to aid individuals in health recovering). These exergames combine therapeutic exercise and gameplay, using technology to transform rehabilitation into an enjoyable and motivating experience. We are working in a comprehensive rehabilitation platform based on mobile exergames for cervical rehabilitation to which we refer as RehbeCa. All the exergames included in the platform are designed by physiotherapists with clinical criteria: they include all types of exercises in a home exercise program (i.e., a set of exercises to perform at home). Within these exergames, there is the Stack exergame that was inspired from the original Tower Bloxx game. This exergame has the aim of encouraging patients to perform neck rotation movements (horizontal movement in the transverse -axial-plane) for cervical rehabilitation purposes, and it is based in the building of two houses stacking blocks alternatively on two bases.

In the course of our research, a question emerged: If the user movements involved to build the stacks follow the same principles than the one-directional point-select proposed by the ISO 9241-411, could this rehabilitation exergame, designed with therapeutic intent, also serve as a testing ground for Fitts’ Law? While Fitts’ Law has wide application in the evaluation of traditional computer devices, to the best of our knowledge, its use in the context of rehabilitation exergames has remained unexplored. Once proved,

this could be a great opportunity to analyze the neck movement performance of different levels of severity of cervicgia¹ and, even, to assess its utility as a diagnosis aid.

In this work, we aim to present a novel integration of rehabilitation and Fitts' law evaluation to provoke the HCI community to look beyond the mere pursuit of technological goals and to exploit every opportunity to deepen into the study of human behavior. Also, we included the analysis of the effect of the sex, to ensure that we detected possible differences between female and male performance, as this is a factor frequently obliterated in the literature. In this study we describe a user study to test if a previously developed rehabilitation exergame could also be used to calculate Fitts' law throughput, with the ultimate goal in mind of testing the potential dual role of rehabilitation exergames as both a therapeutic tool (their original main purpose) and as a HCI tool.

2 REHBECA: THE PLATFORM FOR MOBILE CERVICAL REHABILITATION

RehbeCa is a platform for mobile cervical rehabilitation that emerged based on experiences of collaboration between health and computer scientists [17]. The goal of the platform is to focus on therapeutic exercise, with clinical criteria for user demand and load, for the treatment of nonspecific cervicgia. For that reason, the platform includes a specific mobile application that monitors and analyses the fulfillment of neck therapeutic exercises at home (or elsewhere) [16].

The platform comprises a set of different exergames individually adapted to the user capabilities at any time of the rehabilitation process. At present, it incorporates three exergames:

- (1) **Spaceship**: to train neck movements in the sagittal (longitudinal) plane: neck flexion and neck extension [2].
- (2) **Stacks**: to train neck movements in the transverse (axial) plane: right and left rotation of the neck.
- (3) **Darts**: to train the combination of the three planes of range of movement of the neck, with the six possible movements: flexion, extension, right and left rotation, and right and left lateral flexion, alone or combined [16].

The interaction with the exergames should be performed through the movement of the head of patients. Therefore, the platform integrates a non-invasive mobile camera-based head-tracker. We direct the reader to Roig-Maimó et al. [15] for a technical description of the head-tracker interface.

When a patient logs into the mobile platform, there is an initial step of warming up guided by instructional videos. Then, the patient has access to the exergames designated for his or her profile. Finally, a cool-down movie completes the workout. Additionally, the platform stores the performance data in order to monitor the clinical progress of a patient and to provide clinical proof of the reliability, efficiency and safety of the system [17]. The clinical evaluation of the RehbeCa platform as a therapeutic tool is currently in process and out of the scope of this paper.

¹Cervicgia is the name given to pain that appears in the area of the cervical spine, the posterior part of the neck

3 THE TOWER BLOXX GAME

Tower Bloxx was a mobile phone game released by Digital Chocolate in the early 2000s. The game was quite popular among players since it was one of the classic games on mobile phones at that time. It was a tower building game (see Figure 1a).

In Tower Bloxx, players were challenged with the construction of towering skyscrapers in a urban environment. The primary objective was to create stable and towering buildings by strategically stacking block segments on top of one another. The interface featured a tower base onto which the blocks needed to be stacked. In Figure 1a, you can observe the tower base with one block already in place. The blocks appeared swinging on a construction crane. When the player determined that a block was in an optimal position (typically, directly above the center of the base), the block could be released by pressing a key on the keyboard of the mobile phone. The objective was to construct a tower that was both tall and straight. The precision and timing of each placement were crucial; if a block landed outside the edges of the block below it, it was deemed an error, as this could potentially lead to the collapse of the entire structure. Players were allowed only three errors before the game concluded.

Nowadays, new versions of the classic game can be found online [1] or on mobile application marketplaces [18, 21, 22].

4 THE EXERGAME: THE STACK EXERGAME

The Stack exergame (see Figure 1b-c) is one of the exergames included in the RehbeCa platform. The clinical aim of this exergame is to train the neck rotation movement (horizontal movement in the transverse -axial- plane), as the RehbeCa platform already has exergames to train neck movements in the sagittal plane (flexion and extension) with the Spaceship exergame [2] and the combination of neck movements (flexion, extension, rotation and lateral flexion) with the Dart exergame [16].

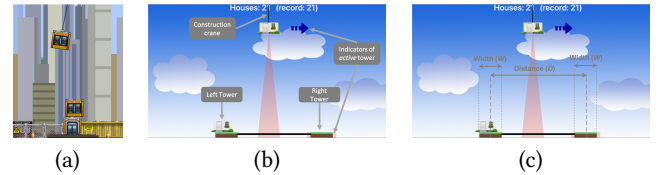


Figure 1: Screenshot of the (a) Tower Bloxx game [4], and screenshots of the Stack exergame included in the RehbeCa platform running on an Apple iPhone: (b) with annotations, and (c) depicting the configuration parameters' of the levels.

The Stack exergame was inspired from the original Tower Bloxx game (see Section 3). With the aim of encouraging patients to perform neck movements in the transverse plane, this game has as a goal the construction of two towers (see "Left Tower" and "Right Tower" in Figure 1b), each complete with its own tower base. This is a departure from the original version, which had only one tower. The goal of the game is to build both towers by stacking blocks alternately on the respective tower bases. During each move, the exergame designates the *active* tower (i.e., the tower where the

current block must be stacked), which is highlighted by a salmon-colored base and a directional blue arrow (see "Indicators of active tower" in Figure 1b).

In the Stack exergame, the position of the construction crane, with a hanging block (see "Construction crane" in Figure 1b), is controlled through the movement of the head of the patient. To release the block, the player simply needs to touch anywhere on the screen.

Physiotherapists must manage the range of motion (ROM) and muscle control (accuracy of movement) of the rotations that patients must perform [10]. To do so, they can configure different levels for each patient with two parameters. First, the distance between the bases of the two towers (see D in Figure 1c) to control of the movement's amplitude (ROM required); and second, the width of the bases of the towers (see W in Figure 1c) to control the accuracy of the movement.

The values of the configuration of the parameters of the levels were stated by the physiotherapists in Salinas-Bueno et al. [16], taking into account their safety in terms of the neck range of motion (ROM) for each direction and movement and were:

- Distance: 280, 560, 1120 px
- Width: 88, 132, 198 px

This configuration generates nine difficulty levels, as a combination of the levels of the two configuration parameters'. The difficulty of the levels was stated following the criteria that a longer distance implies a higher difficulty, and less width also implies more difficulty. The difficulty of each level was calculated applying the Shannon formulation of the index of difficulty (ID) of Fitts' Law [20]:

$$ID = \log_2 \left(\frac{D}{W} + 1 \right)$$

For safety and rehabilitation purposes, patients must perform through the levels in ascending difficulty order (see Table 1).

Table 1: Description of the levels of the game in ascending difficulty order

Level	Distance (px)	Width (px)	ID
1	280	198	1.27
2	280	132	1.64
3	560	198	1.94
4	280	88	2.06
5	560	132	2.39
6	1120	198	2.73
7	560	88	2.88
8	1120	132	3.25
9	1120	88	3.78

In each level, patients have to place a total of 25 blocks (both towers included). If a block is misplaced, the information is recorded but there is no penalty.

5 ISO TESTING AND THE CALCULATION OF THROUGHPUT

ISO 9241-411 [7]² describes performance tests for evaluating the efficiency and effectiveness of existing or new non-keyboard input devices. Specifically, the ISO standard proposes a one-directional point-select test and a multi-directional target-select test (see Figure 2).

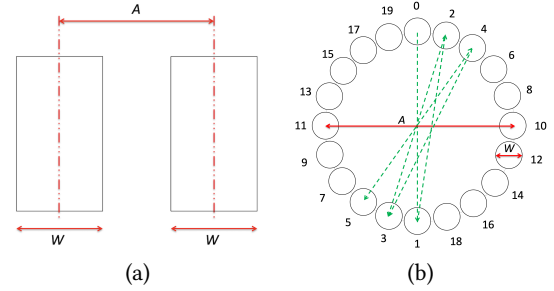


Figure 2: ISO tests for pointing evaluation: (a) one-directional point-select test, and (b) Multi-directional target-select test (numbering indicates the order of target selection)

As stated by the ISO "input devices should be tested for the tasks for which they are intended to be used". Consequently, as most of pointing devices are for multi-directional pointing, the multi-directional tapping test usually is the test used as a standard *de facto*. However, a significant difference on throughput between the two tests does not always occur [14] with multi-directional pointing interfaces.

In particular, the one-directional point-select test (see Figure 2a), inspired by the original Fitts paradigm, is very similar to the interaction performed in the Stack exergame: moving continuously back and forward between two targets.

Therefore, taking into account all the above, in this research we focus on the one-directional point-select test.

The one-directional point-select test evaluates pointing movements along one axis. The participant is required to move a cursor between two targets of width W separated by a centre-to-centre amplitude A a predefined number of times (see Figure 2a).

The ISO standard specifies throughput (TP) as the performance measure (in bits per second) and it is calculated using the formulation:

$$TP = \frac{\text{Effective index of difficulty}}{\text{Movement time}} = \frac{ID_e}{MT'}$$

where ID_e is computed from the movement amplitude (A) and target width (W) and MT is the per-trial movement time averaged over a sequence of trials.

A detailed description of the calculation of throughput can be found in other sources [11, 19, 20].

²ISO 9241-411 [7] is an updated version of ISO 9241-9 [6]. With respect to performance evaluation, the two versions of the standard tests are the same.

6 THE EXPERIMENT

In this section we present the experiment to evaluate if the Stack exergame could be used as an alternative to the ISO one-directional point-select test to calculate Fitts' law throughput. Therefore, the main goal of the experiment is to evaluate the effect of the two apparatus on throughput. The first apparatus implements the one-directional point-select test proposed by the ISO (see Figure 3a). The second apparatus implements the task of the Stack exergame (see Figure 3b), which is very similar to the task of the ISO test but driven by a game-like experience.

6.1 Participants

Following the Sex and Gender Equity in Research – SAGER – guidelines [5], we recruited a gender balanced sample of fourteen participants (7 females) from community centers in local towns and university campus staff and students. Ages ranged from 10 to 56 with a mean of 36.3 years ($SD = 15.0$). There were no requirements on prior experience to participate in the study. None of the participants had previous experience with any of the apparatus used in the experiment.

This balanced set of participants allowed to include the sex factor in the analysis, as differences on throughput by sex for mobile head-based interaction were expected [13].

6.2 Apparatus

The experiment was conducted on an Apple *iPhone 7 Plus* with a resolution of 1920×1080 pixels and a pixel density of 401 ppi. All communication with the mobile device was disabled during testing.

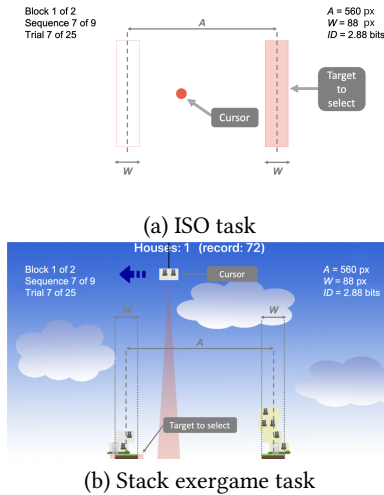


Figure 3: Screenshots of the experiment software: example target condition with annotations ($A = 560$ px, $W = 88$ px) for the (a) ISO apparatus and the (b) Stack exergame apparatus.

We implemented two different apparatus that follow the original Fitts paradigm: the ISO one-directional test (see Figure 3a) and the Stack exergame (see Figure 3b).

For both apparatus, each sequence consisted of 25 targets with the target to select highlighted for each trial. Upon selection, a new target was highlighted. The target was highlighted in green when the cursor was inside it.

In the ISO apparatus we implemented the one-directional point-select test (see Figure 2a and Figure 3a). In the Stack exergame apparatus (see Figure 3b), the participant has to build towers by stacking blocks. In that case, the targets to select are translated as the towers' bases and the cursor is translated as the construction crane holding the block that has to be released. The user controls the horizontal position of the construction crane by the pointing action. When the block is released (by the selection action), it falls to the ground in the same horizontal position where it has been released. Using the same philosophy than the ISO one-directional test, the participant has to build two towers simultaneously and alternately. While in the ISO apparatus the index of difficulty of the current task is randomized within blocks, in the Stack exergame apparatus, as already commented in Section 4, we use a progressive index of difficulty due to rehabilitation purposes.

For both of them, the user input combined mobile head-tracking for the pointing action and touch for selection.

6.3 Procedure

The experiment was conducted according to the guidelines of the Declaration of Helsinki, and previously approved by the Research Ethics Committee of our university (Exp.223CER21). After signing a consent form, participants were briefed on the goals of the experiment and were instructed to sit and hold the device in landscape orientation in a comfortable position. The only requirement was that their entire face was visible by the front camera of the device.

The experiment task was demonstrated to participants, after which they did a few practice sequences. They were instructed to move the cursor by holding the device still and moving their head. Selection occurred by tapping anywhere on the display surface with a thumb. Testing began after they felt comfortable with the task and the interaction method. The test lasted about half an hour per participant and participants were allowed to rest as needed between sequences.

6.4 Design

The experiment was fully within-subjects with the following independent variables and levels:

- Apparatus: ISO one-directional (ISO), Stack exergame (Exergame)
- Block: 1, 2
- Amplitude: 280, 560, 1120 px
- Width: 88, 132, 198 px

The primary independent variable was the apparatus. Block, amplitude, and width were included to gather a sufficient quantity of data over a reasonable range of task difficulties (with ID s ranging from 1.27 to 3.78 bits). Although this range of task ID s does not cover the range of [2 - 8] recommended by Soukoreff and MacKenzie [20], it lies within the expected range of task ID s when dealing with devices with limited size of the display, like mobile devices, due to their limited movement amplitudes between targets [12].

For each condition, participants performed a sequence of 25 trials. The two apparatus were assigned using a 2×2 Latin square with 7 participants per group. For the ISO apparatus, the amplitude and width conditions were randomized within blocks. For the Stack exergame apparatus, the amplitude and width conditions were presented in an increasing order of difficulty within blocks.

The dependent variables were throughput, movement time, and error rate.

The total number of trials was 14 participants × 2 tasks × 2 blocks × 3 amplitudes × 3 widths × 25 trials = 12,600.

Following the Sex and Gender Equity in Research – SAGER – guidelines [5], we also investigated if there was any difference by sex over the dependent variables.

6.5 Results

In this section, we present results for throughput, movement time, and error rate. Initially, we assess the impact of the within-subjects factor *Apparatus*, and subsequently, we examine the influence of the *Sex* factor on all the dependent variables.

6.5.1 Throughput. Figure 4a shows the results obtained for throughput by apparatus and sex. The grand mean for throughput was 1.03 bps. This value is low compared to a desktop mouse where throughput is typically about 4.5 bps [20], but closer to the value of throughput for mobile tilt-based interaction (around 1.75 bps [8]) and to head input on desktop environments (from 1.28 bps to 2.10 bps [3, 9]).

The mean throughput for the ISO apparatus was 1.11 bps, which was 16.8% higher than the mean throughput of 0.95 bps for the Stack exergame apparatus. The difference was not statistically significant ($F_{1,13} = 4.19, p > .05$).

Applying a between-subjects design where the independent variable was *Sex* with the levels *female* and *male*, the difference on the global throughput was statistically significant ($F_{1,502} = 18.47, p < .001$). The mean throughput obtained by the female participants was 0.94 bps, which was 16.1% lower than the mean throughput of 1.12 bps obtained by the male participants. The difference observed on throughput by sex was expected due to previous work [13].

Analyzing the effect of sex on throughput by apparatus, for the male participants the difference was not statistically significant ($F_{1,6} = 0.66, ns$). For the female participants, the difference was statistically significant ($F_{1,6} = 12.7, p < .01$): the mean throughput for the ISO apparatus was 1.07 bps, which was 32.1% higher than the mean throughput of 0.81 bps for the Stack exergame apparatus.

6.5.2 Movement Time and Error Rate. Figure 4b shows the results obtained for movement time by apparatus and sex. The grand mean for movement time was 1.98 s per trial. By apparatus, the means were 1.78 s (ISO) and 2.18 s (Exergame). The difference was statistically significant ($F_{1,13} = 13.17, p < .01$).

Analyzing the effect of sex on movement time per trial, the difference on the global movement time was statistically significant ($F_{1,502} = 18.8, p < .001$). The mean movement time per trial obtained by the female participants was 2.12 s, while the mean movement time obtained by the male participants was 1.82 s. By apparatus, for the male participants the difference was not statistically significant ($F_{1,6} = 1.47, p > .05$). For the female participants,

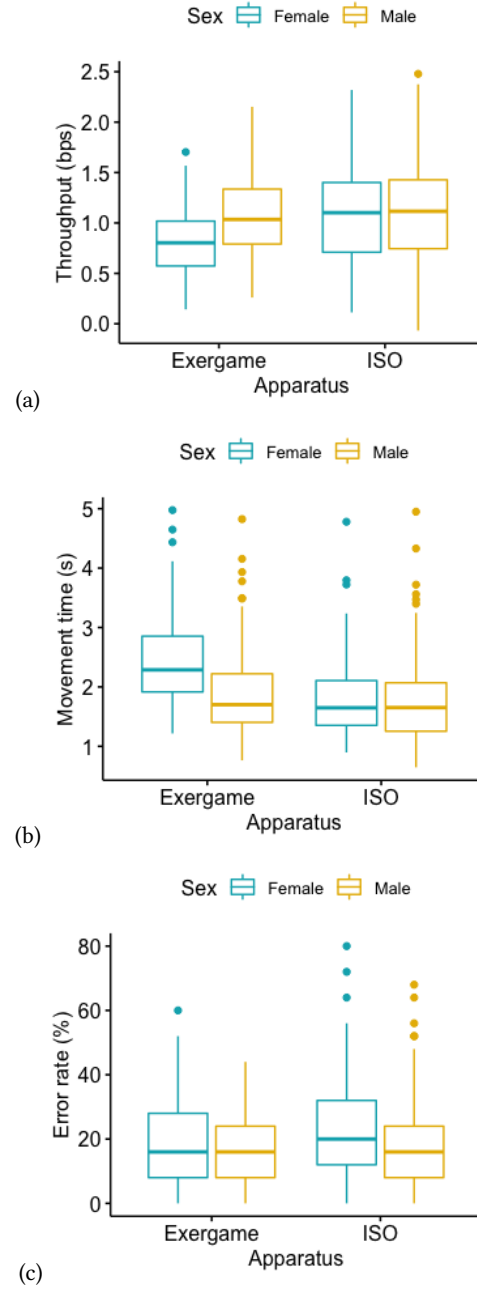


Figure 4: Throughput (bps), movement time (s), and error rate (%) by apparatus and sex.

the difference was statistically significant ($F_{1,6} = 25.83, p < .01$): the mean movement time per trial for the ISO apparatus was 1.80 s, which was 26.23% faster than the mean movement time per trial of 2.44 s for the Stack exergame apparatus.

Figure 4c shows the results obtained for error rate by apparatus and sex. The grand mean for error rate was 18.98% per sequence. By

apparatus, the means were 20% (ISO) and 17.95% (Exergame). The difference was not statistically significant ($F_{1,13} = 1.30, p > .05$).

Analyzing the effect of sex on error rate, the difference on the global error rate was statistically significant ($F_{1,502} = 5.38, p < .05$). The mean error rate obtained by the female participants was 20.4%, while the mean error rate obtained by the male participants was 17.6%. By apparatus, the effect of sex on error rate was not statistically significant neither for females ($F_{1,6} = 1.45, p > .05$) nor for males ($F_{1,6} = 0.07, ns$).

7 DISCUSSION AND CONCLUSIONS

Our study initially revolved around the domain of cervical rehabilitation exergames within the broader context of health tech, aiming to leverage technology for therapeutic purposes. As we ventured deeper into this domain, we recognized a unique opportunity to extend the application of one of these exergames. Beyond their role in rehabilitation, we saw its potential to serve as a testing apparatus for Fitts' Law (as its task is very similar to the task of the one-directional point-select test proposed by the ISO 9241-411).

We conducted a user study to investigate the influence of apparatus (the Stack exergame and the ISO one-directional) on throughput, movement time, and error rate; analyzing also the effect of the sex factor.

In terms of throughput, we found that the effect of the apparatus was not statistically significant. This result confirms that the gamified elements and complexities inherent to the stack exergame, do not introduce biases nor deviate from traditional Fitts' law testing conditions potentially impacting the accuracy of the results of throughput. Therefore, the stack exergame can be used for Fitts' law evaluation. This opens the door to investigate further into two key areas: (1) calculate the throughput of the patients while doing their exercises of rehabilitation with the exergame; which would allow to obtain benchmark values of neck performance for different levels of severity. (2) Investigate if the value of Fitts' law throughput could serve as an alternative diagnosis measure to classify patients of cervicgia. Despite in global there was no difference of throughput by apparatus, when we analyzed the data separately by sex, we found a significant difference by apparatus with the females participants. This finding needs further research as may have a high impact on the design of exergames (and, by extension, of games). Which is the reason behind the fact that the throughput obtained by the females participants is lower in a gamified apparatus than in an apparatus with a simpler interface, while this effect does not occur in male participants?

Aligning with previous research [13], we also found a difference of throughput by sex. This result highlights the importance of sex-balanced participants and the ulterior analysis of the sex factor. Future research should also dig out about the reason behind why the throughput obtained by the females participants using a head-tracker is lower than the male participants.

In this work, we explored the feasibility of a novel integration of rehabilitation and HCI evaluation, unlocking possibilities that could inspire further research and practical applications in HCI in the healthcare domain. Starting as an exergame with rehabilitation purposes, our research has shown how this technology could evolve also into a HCI tool, marking a significant step forward in advancing

the field. Specifically, we described and tested the potential dual role of a therapeutic tool as a platform for studying Fitts' law, with the ultimate future goal of be able to explore the use of Fitts' law throughput for therapeutic diagnosis; closing the cycle.

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REFERENCES

- [1] 2019. *Tower Game*. <https://www.towergame.app>
- [2] Antonio Arenas, Maria Francesca Roig-Maimó, Iosune Salinas-Bueno, Javier Varona, Katia San-Sebastián-Fernández, and Cristina Manresa-Yee. 2019. Therapeutic Exercise Based on Videogames to Improve Neck Pain. In *Proceedings of the XX International Conference on Human Computer Interaction* (Donostia, Gipuzkoa, Spain) (*Interacción '19*). Association for Computing Machinery, New York, NY, USA, Article 61, 2 pages. <https://doi.org/10.1145/3335595.3335599>
- [3] Gamhewage C De Silva, Michael J Lyons, Shinjiro Kawato, and Nobuji Tetsutani. 2003. Human factors evaluation of a vision-based facial gesture interface. In *2003 Conference on Computer Vision and Pattern Recognition Workshop*, Vol. 5. IEEE, 52–52.
- [4] Digital Chocolate, Inc. 2006. *Tower Bloxx*. https://archive.org/details/towerbloxx_en_fr_it_de_es_pt_el_7650_v1310
- [5] Shirin Heidari, Thomas F Babor, Paola De Castro, Sera Tort, and Mirjam Curno. 2016. Sex and Gender Equity in Research: rationale for the SAGER guidelines and recommended use. *Research Integrity and Peer Review* 1, 1 (May 2016), 2.
- [6] ISO. 2002. 9241–9. 2000. Ergonomics requirements for office work with visual display terminals (VDTs) – Part 9: Requirements for non-keyboard input devices. *International Organization for Standardization* (2002).
- [7] ISO. 2012. 9241–411. 2012. Ergonomics of human-system interaction – Part 411: Evaluation methods for the design of physical input devices. *International Organization for Standardization* (2012).
- [8] I Scott MacKenzie and Robert J Teather. 2012. FittsTilt: The application of Fitts' law to tilt-based interaction. In *Proceedings of the 7th Nordic Conference on Human-Computer Interaction: Making Sense Through Design*. 568–577.
- [9] John Magee, Torsten Felzer, and I Scott MacKenzie. 2015. Camera Mouse+ ClickerAID: Dwell vs. single-muscle click actuation in mouse-replacement interfaces. In *Universal Access in Human-Computer Interaction. Access to Today's Technologies: 9th International Conference, UAHCI 2015, Held as Part of HCI International 2015, Los Angeles, CA, USA, August 2-7, 2015, Proceedings, Part I* 9. Springer, 74–84.
- [10] Carmen Martín-Gómez, Rebeca Sestelo-Díaz, Víctor Carrillo-Sanjuan, Marcos José Navarro-Santana, Judit Bardon-Romero, and Gustavo Plaza-Manzano. 2019. Motor control using cranio-cervical flexion exercises versus other treatments for non-specific chronic neck pain: a systematic review and meta-analysis. *Musculoskeletal Science and Practice* 42 (2019), 52–59.
- [11] Maria Francesca Roig-Maimó, I. Scott MacKenzie, Cristina Manresa-Yee, and Javier Varona. 2017. Evaluating Fitts' Law Performance with a Non-ISO Task. In *Proceedings of the XVIII International Conference on Human Computer Interaction* (Cancun, Mexico) (*Interacción '17*). Association for Computing Machinery, New York, NY, USA, Article 5, 8 pages. <https://doi.org/10.1145/3123818.3123827>
- [12] Maria Francesca Roig-Maimó, I. Scott MacKenzie, Cristina Manresa-Yee, and Javier Varona. 2018. Head-tracking interfaces on mobile devices: Evaluation using Fitts' law and a new multi-directional corner task for small displays. *International Journal of Human-Computer Studies* 112 (2018), 1 – 15. <https://doi.org/10.1016/j.ijhcs.2017.12.003>
- [13] Maria Francesca Roig-Maimó and Ramon Mas-Sansó. 2019. The Female Effect: The Case of Gender Parity on User Studies. In *Proceedings of the XX International Conference on Human Computer Interaction* (Donostia, Gipuzkoa, Spain) (*Interacción '19*). ACM, New York, NY, USA, Article 46, 6 pages. <https://doi.org/10.1145/3335595.3335606>
- [14] Maria Francesca Roig-Maimó and Ramon Mas-Sansó. 2024. Is It Worth the Hassle? Moving from One-directional to Multi-directional on Fitts' Law Evaluation. In *Proceedings of the XXIII International Conference on Human Computer Interaction* (Lleida, Spain) (*Interacción '23*). Association for Computing Machinery, New York, NY, USA, Article 3, 4 pages. <https://doi.org/10.1145/3612783.3612787>
- [15] Maria Francesca Roig-Maimó, Cristina Manresa-Yee, and Javier Varona. 2016. A Robust Camera-Based Interface for Mobile Entertainment. *Sensors* 16, 2 (2016).

- <https://doi.org/10.3390/s16020254>
- [16] Iosune Salinas-Bueno, Maria Francesca Roig-Maimó, Pau Martínez-Bueso, Katia San-Sebastián-Fernández, Javier Varona, and Ramon Mas-Sansó. 2021. Camera-Based Monitoring of Neck Movements for Cervical Rehabilitation Mobile Applications. *Sensors* 21, 6 (2021). <https://doi.org/10.3390/s21062237>
- [17] Katia San-Sebastián-Fernández, Javier Varona, Maria Francesca Roig-Maimó, and Iosune Salinas-Bueno. 2018. Clinical evaluation of a mobile app for therapeutic exercise for neck pain. In *Proc. of the Second International Conference on Accessibility, Inclusion and Rehabilitation using Information Technologies* (Palma de Mallorca, Spain). 11–13.
- [18] Muneer Shehat. 2022. *Tower Bloxx 3D*. <https://play.google.com/store/apps/details?id=com.muneershehat.towerbloxx3d>
- [19] I. Scott MacKenzie. 2015. Fitts' Throughput and the Remarkable Case of Touch-Based Target Selection. In *Human-Computer Interaction: Interaction Technologies*, Masaaki Kurosu (Ed.). Springer International Publishing, Cham, 238–249.
- [20] R. William Soukoreff and I. Scott MacKenzie. 2004. Towards a standard for pointing device evaluation: Perspectives on 27 years of Fitts' law research in HCI. *International Journal of Human-Computer Studies* 61, 6 (2004), 751–789. <https://doi.org/10.1016/j.ijhcs.2004.09.001> Fitts' law 50 years later: applications and contributions from human-computer interaction.
- [21] Tap by Tap, LLC. 2018. *Tower Blocks Puzzle: Craft It*. <https://apps.apple.com/us/app/tower-blocks-puzzle-craft-it/id659265195>
- [22] ZeekGame. 2023. *City Bloxx*. <https://play.google.com/store/apps/details?id=com.zeek.citybloxx>